SUPPORTING ADULT GUIDANCE

Fractions, decimals and percentages fluently

Number Systems and Calculator Judgement · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This opening lesson of Number Systems establishes fluency across fractions, decimals and percentages, and with the four operations on fractions. Learners move a single point on a number line and watch its three forms change together, use area and bar pictures for fraction calculations, order numbers given in mixed forms, choose the clearest form for real situations, and solve fraction-of-amount and find-the-whole problems.

Driving Question: *How many ways can one number be written?*

The mathematics, in plain terms

A rational number has equivalent representations as a fraction, a decimal and a percentage. A fraction in its simplest form terminates as a decimal exactly when its denominator has no prime factors other than 2 and 5; otherwise the decimal recurs. Addition requires a common denominator; multiplication multiplies numerators and denominators; division by a fraction is multiplication by its reciprocal.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Three names for one number

Write 0.35 as a fraction in its simplest form, and as a percentage.

1. 0.35 is 35 hundredths.	35/100
2. Divide the numerator and the denominator by 5.	7/20
3. Multiply the decimal by 100 for a percentage.	35%

0.35 = 7/20 = 35%.

Example 2 — Dividing by a fraction: from the picture to the method

Work out 2 1/2 ÷ 3/4.

1. Dividing asks how many 3/4 pieces fit into 2 1/2. Draw a bar and count.	3 pieces make 2 1/4; 1/3 of a piece more makes 2 1/2
2. So the answer is 3 and a third of a piece.	3 1/3
3. Why the method works: each whole holds 1 1/3 = 4/3 pieces of 3/4.	1 ÷ 3/4 = 4/3
4. So 2 1/2 wholes hold 2 1/2 × 4/3 pieces. Multiply.	5/2 × 4/3 = 20/6 = 10/3 = 3 1/3 ✓

2 1/2 ÷ 3/4 = 3 1/3. Dividing by 3/4 is the same as multiplying by its reciprocal, 4/3.

Size before method

Encourage learners to say roughly where a number sits before converting or calculating. In Fraction pictures, each picture leads to its method: the area model shows why numerators and denominators are multiplied, and 'how many fit?' shows why dividing by 3/4 is multiplying by 4/3. Let learners see the picture before naming the rule.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The linked number line: one point, three names; place, convert and order.	~14 min
Movement	Fraction pictures: area and bar models for adding, multiplying and dividing.	~10 min
Reflection	Which form wins? Order mixed forms, then choose the clearest form for real situations.	~7 min
Creativity	Fractions of amounts, one amount as a fraction of another, and finding the whole.	~9 min
Rest	Half a pizza, 0.5 of a pizza, 50% of a pizza: the same amount.	~2 min
Nutrition	Intellectual: number sense — seeing the size behind the form.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding numerators and denominators (1/3 + 1/4 = 2/7).
Ask: is the answer bigger than 1/3 on its own? Then cut both into twelfths.

Thinking 0.38 is bigger than 0.4 because 38 is bigger than 4.
Write both to the same number of places: 0.38 and 0.40.

Dividing by a fraction gives a smaller answer.
Ask: how many 3/4s fit into 3? More than 3.

In 'find the whole' problems, taking the fraction of the part (3/4 of 90).
Draw the bar: 90 is three of the four parts.

Comparing a change with the new value instead of the original.
Ask: what was it before the change?

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: equivalent fractions and adding fractions. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 3/8 is smaller than 35% because 3 is smaller than 35. The aim is a question that lets them find the slip themselves.

- "Where does 3/8 sit on the number line? And 35%?"
- "What is 3/8 as a decimal?"

- “What is 35% as a decimal?”
- “Which form makes these two easiest to compare?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are calculators used?
Not in this lesson: the aim is understanding the forms. Calculators are used deliberately in Lesson 3.

Why accept 0.583 for 7/12?
A correctly rounded decimal shows the right value. Exact forms are expected where the question asks for a fraction.

What about recurring decimals?
They are met here (2/3) and are the focus of Lesson 2, including converting them back to fractions.

What comes next?
Lesson 2: recurring decimals as fractions.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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